

Lab 2 — Agile Backlog Creation & Sprint Simulation in Jira

Problem Statement #24 — Public Bus Live Tracking & Crowding Estimator

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Course	Software Engineering Lab — PES University, Dept. of CSE
Jira Project	Public Bus Live Tracking (key SCRUM)
Jira Workspace Link	https://stu-team-zdgyycel.atlassian.net/jira/software/projects/SCRUM/boards/1
GitHub Repository	https://github.com/LOKESH-ADITHYA/SE-Labs-PES1UG24CS392

Summary

Item	Value
Epics	5 — one per Lab 1 functional requirement (FR-001 ... FR-005)
User Stories	14, all in <i>As a / I want / So that</i> format with acceptance criteria
Story Points	91 total, Fibonacci scale (3, 5, 8, 13)
Priorities	High / Medium assigned to every story
Sprint 1	18 – 25 Aug 2026 · 42 points committed · 42 completed · Closed
Sprint 2	25 Aug – 1 Sep 2026 · 49 points committed · 36 completed · Closed

1. Epics and User Stories

The screenshot displays the Jira 'All work' view for the 'Public Bus Live Tracking' project. The interface shows a list of 19 work items, categorized into 5 Epics (SCRUM-1 to SCRUM-5) and 14 User Stories (SCRUM-6 to SCRUM-19). Each item is listed with its title, assignee, reporter, priority, and status. The Epics are all 'To Do' with Medium priority. The User Stories are all 'Done' with High or Medium priority. The interface includes a sidebar with navigation options like 'For you', 'Recent', 'Starrd', 'Apps', 'Plans', 'Spaces', 'Filters', and 'Dashboards'. The top bar shows search, create, and plan buttons.

Work	Assignee	Reporter	Priority	Status
SCRUM-1 EPIC 1: Real-Time Vehicle Tracking & ETA Engine	Unassigned	S.R Lokesh Adit...	Medium	To Do
SCRUM-2 EPIC 2: Passenger Crowding Estimation	Unassigned	S.R Lokesh Adit...	Medium	To Do
SCRUM-3 EPIC 3: Commuter Journey Information	Unassigned	S.R Lokesh Adit...	Medium	To Do
SCRUM-4 EPIC 4: Arrival Alerts & Notifications	Unassigned	S.R Lokesh Adit...	Medium	To Do
SCRUM-5 EPIC 5: Fleet Operations Control	Unassigned	S.R Lokesh Adit...	Medium	To Do
SCRUM-6 Story 1.1: GPS Telemetry Ingestion	Unassigned	S.R Lokesh Adit...	High	Done
SCRUM-7 Story 1.2: ETA Calculation Engine	Unassigned	S.R Lokesh Adit...	High	Done
SCRUM-8 Story 1.3: Stale Telemetry Falback	Unassigned	S.R Lokesh Adit...	Medium	Done
SCRUM-9 Story 2.1: Ticketing Sensor Data Capture	Unassigned	S.R Lokesh Adit...	High	Done
SCRUM-10 Story 2.2: Occupancy Calculation and Crowd...	Unassigned	S.R Lokesh Adit...	High	Done

All 19 work items in the project — 5 Epics (SCRUM-1 to SCRUM-5) and 14 User Stories (SCRUM-6 to SCRUM-19), each with its priority and status.

2. Backlog with Story Points

Public Bus Live Tracking

Summary List Board **Backlog** Development Forms More 3 +

Search backlog Filter

SCRUM Sprint 2 25 Aug – 1 Sep (8 work items) 8 / 5 / 36 Complete sprint ...

Deliver the user-facing layer: commuter journey information, arrival alerts and fleet operations control.

ID	Story Name	Parent Epic	Status	Story Points	Assignee
SCRUM-12	Story 3.1: Route and Stop Search	EPIC 3: C...	Done	5	
SCRUM-13	Story 3.2: Live Bus Position Map	EPIC 3: C...	Done	8	
SCRUM-14	Story 3.3: Arrival List with ETA and...	EPIC 3: C...	Done	5	
SCRUM-15	Story 4.1: Subscribe to Arrival Alert	EPIC 4: A...	Done	5	
SCRUM-16	Story 4.2: Push Notification Delivery	EPIC 4: A...	Done	5	
SCRUM-17	Story 5.1: Fleet Monitoring Dashbo...	EPIC 5: Fl...	Done	8	
SCRUM-18	Story 5.2: Automatic Exception Fla...	EPIC 5: Fl...	In Progress	5	
SCRUM-19	Story 5.3: Dispatch Relief Bus	EPIC 5: Fl...	To Do	8	

+ Create

Backlog (0 work items) 0 / 0 / 0 Create sprint

Your backlog is empty.

Sprint 2 backlog view showing each User Story with its parent Epic, status and story-point estimate (8, 5, 5, 5, 5, 8, 5, 8). The sprint header shows the 8 / 5 / 36 status split and the sprint goal.

3. Active Sprint Board

Public Bus Live Tracking

Summary List **Board** Backlog Development Forms Timeline Docs Reports +

Search board Filter Group

Complete sprint

To Do 1

Story 5.3: Dispatch Relief Bus
Parent: EPIC 5: Fleet Operations Control
SCRUM-19 8

In Progress 1

Story 5.2: Automatic Exception Flagging
Parent: EPIC 5: Fleet Operations Control
SCRUM-18 5

In Review 0

+ Create

Done 6

Story 3.1: Route and Stop Search
Parent: EPIC 3: Commuter Journey Inf...
SCRUM-12 5 ✓

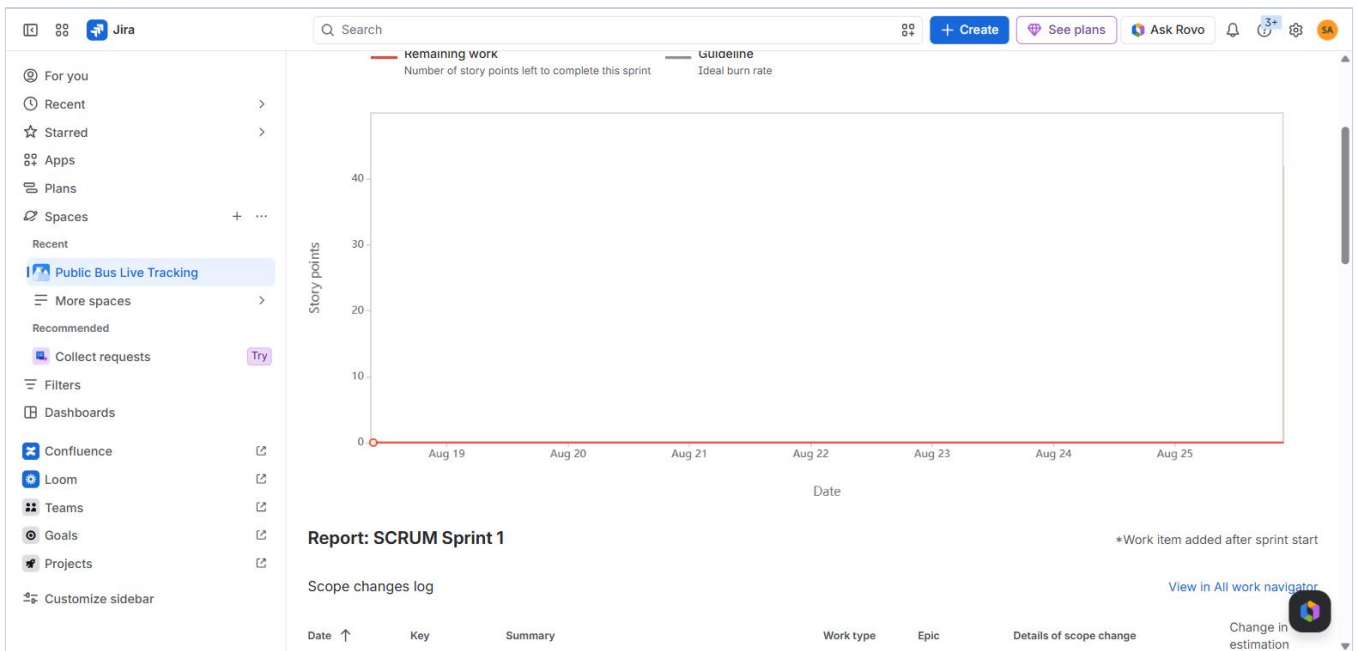
Story 3.2: Live Bus Position Map
Parent: EPIC 3: Commuter Journey Inf...
SCRUM-13 8 ✓

Story 3.3: Arrival List with ETA and Crowding Badge
Parent: ...
SCRUM-14 5 ✓

+ Create

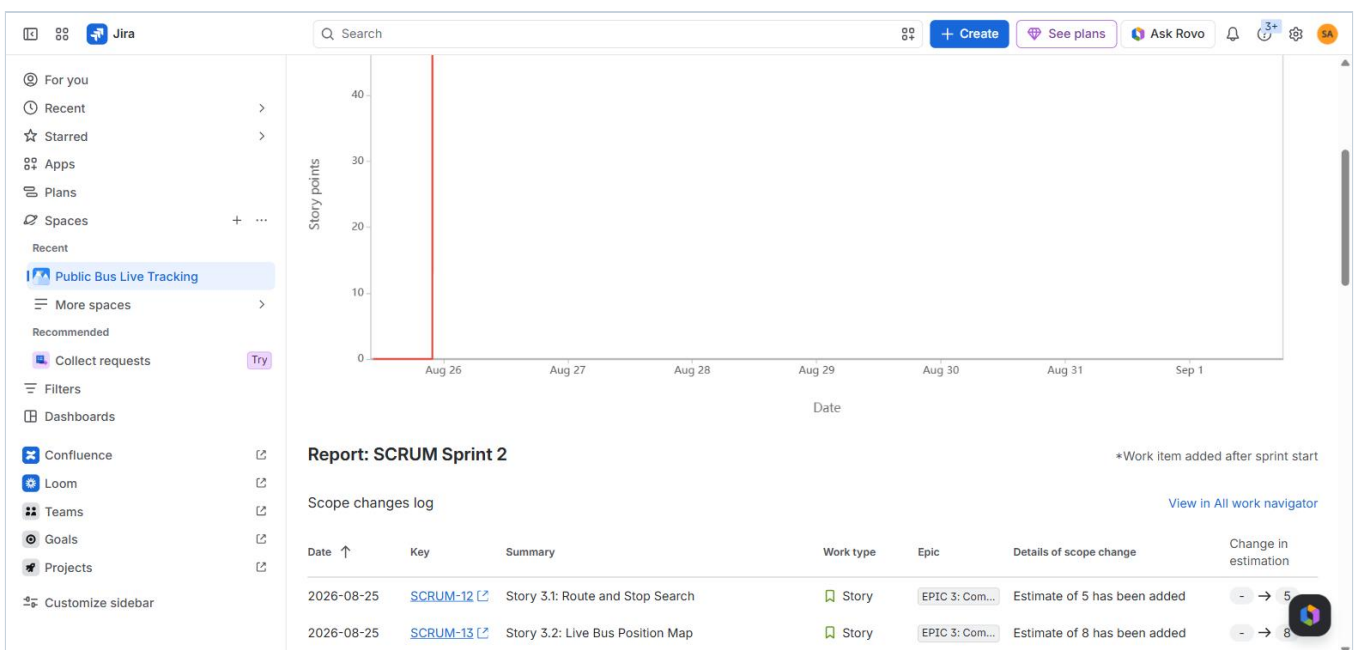
Sprint 2 board mid-sprint: To Do (1), In Progress (1), In Review (0), Done (6) — showing stories progressing across the workflow columns.

4. Burndown Chart — Sprint 1



Sprint 1 (18 – 25 Aug), 42 story points, all completed. The vertical shape reflects the simulation: all status changes were made in one session rather than spread across the sprint window.

5. Burndown Chart — Sprint 2



Sprint 2 (25 Aug – 1 Sep), 49 story points committed, 36 completed. The remaining 13 points (SCRUM-18, SCRUM-19) left the sprint as a scope reduction at close, not as delivered work.

Reflection

Backlog summary

Epic	Traces to	Stories	Points
EPIC 1 — Real-Time Vehicle Tracking & ETA Engine	FR-001	3	26
EPIC 2 — Passenger Crowding Estimation	FR-002	3	16
EPIC 3 — Commuter Journey Information	FR-003	3	18
EPIC 4 — Arrival Alerts & Notifications	FR-004	2	10
EPIC 5 — Fleet Operations Control	FR-005	3	21
Total		14	91

Sprint outcomes

	Sprint 1	Sprint 2
Dates	18 – 25 Aug (1 week)	25 Aug – 1 Sep (1 week)
Scope	EPIC 1 + EPIC 2 (SCRUM-6 → SCRUM-11)	EPIC 3, 4, 5 (SCRUM-12 → SCRUM-19)
Committed	42 points, 6 stories	49 points, 8 stories
Completed	42 points (100%)	36 points (73%)
Carried over	none	SCRUM-18 (5 pts), SCRUM-19 (8 pts) — returned to backlog at close
Status	Closed	Closed

1. Did your estimations reflect the actual effort?

Partly, and the exercise exposed where estimation is weakest.

Story 1.2 *ETA Calculation Engine* was pointed at **13** — the largest item in the backlog — because it carries the most uncertainty rather than the most typing. Distance-to-time conversion depends on live traffic speed, and the acceptance criterion attached to it (p95 recomputation under 2 seconds across 1,000 vehicles, from NFR-001) is a performance target, not a feature. Under planning poker this is exactly the item where estimates would diverge most, because the effort depends on how well the traffic model behaves, which nobody knows until it is built.

By contrast Story 2.3 *Crowding Refresh on Stop Departure* was pointed at **3** and is genuinely small: it re-runs an existing calculation on an event that already exists.

The estimates that proved least reliable were in Sprint 2. *Dispatch Relief Bus* (SCRUM-19, 8 points) never started, and *Automatic Exception Flagging* (SCRUM-18, 5 points) did not finish. Both sit at the end of a dependency chain — neither can begin until the Fleet Dashboard exists — so their true cost includes waiting, which story points do not capture at all. Points measure the size of the work, not when it can be started.

The honest limitation is that this was a **simulated** sprint. Cards were moved To Do → In Progress → Done without real implementation, so there is no measured effort to compare against. What the exercise did validate is the *relative* ordering, which is what story points are actually for. The Fibonacci gaps helped: the jump from 8 to 13 forced a decision about whether the ETA engine was "hard" or "much harder" rather than allowing a comfortable 10.

2. Was your backlog well-prioritised?

Yes, and the priorities follow a defensible rule rather than instinct: **a story is High if something else in the backlog is unusable without it.**

- FR-001 and FR-002 work (Stories 1.1, 1.2, 2.1, 2.2) is High because every commuter-facing feature consumes ETA and crowding data. Nothing else can be demonstrated until these exist.
- The commuter query path (Stories 3.1, 3.2, 3.3) is High because without it the computed data never reaches a user — the system would be correct and invisible.
- Alerts (EPIC 4) are Medium. They are convenience layered on a system that already works. This matches Lab 1, where FR-004 is Priority Medium and *Subscribe to Arrival Alert* is modelled as an «**extend**» relationship rather than an «include» — optional behaviour, by definition.
- Story 1.3 *State Telemetry Fallback* is Medium despite being the most interesting design decision in the project, because the system delivers value before it is added.

The prioritisation is validated by what actually got dropped. The two stories that slipped out of Sprint 2 were both Medium-priority items at the end of the dependency chain — exactly the ones a well-ordered backlog *should* shed when a sprint runs short. Nothing High-priority was left unfinished in either sprint, which is the real test.

The one debatable call is Story 5.3 *Dispatch Relief Bus* at Medium. It is the only action in the whole system that changes the physical world, which argues for High. It was set Medium because it cannot start until Stories 5.1 and 5.2 are done, so raising it would not have changed the order — and indeed it is the story that did not get started.

3. How did your simulated sprint align with your plan?

The two sprints diverged, and the difference is the most useful thing in this lab.

Sprint 1 landed exactly on plan: 42 points committed, 42 delivered, nothing carried over. Every story in it belonged to the data layer and had no external dependency, so there was nothing to block.

Sprint 2 missed: 49 points committed, 36 delivered — 73%. When the sprint was closed, *Automatic Exception Flagging* (SCRUM-18, 5 pts, still In Progress) and *Dispatch Relief Bus* (SCRUM-19, 8 pts, never started) were incomplete and returned to the backlog. The cause is visible in the plan rather than in execution. Sprint 2 was given 49 points on the strength of Sprint 1's 42, but Sprint 1's velocity was measured on independent work, while Sprint 2 stacked EPIC 5 on a chain where *Dispatch Relief Bus* waits on *Exception Flagging*, which waits on the *Fleet Dashboard*. Sequential work cannot be parallelised however many points are available.

The lesson is that **committing to a sprint on last sprint's point total ignores the shape of the work.** Two sprints of identical size can have very different capacity depending on how much of the work can proceed at once. A fairer Sprint 2 would have been around 40 points, moving *Dispatch Relief Bus* to a third sprint from the start rather than discovering it at the end.

A caveat on realism: 42 points in one week implies roughly 45 points per developer-week, which is not achievable — a 13-point story alone maps to three or four days on the handout's own conversion table. The point totals are internally consistent for relative estimation but are not calibrated against real capacity.

4. What insights did the burndown chart give about your team's capacity?

The two charts have to be read with one caveat stated up front: because this is a **simulation**, every status change was made in a single working session rather than spread across the sprint window. Both charts therefore fall as a near-vertical line rather than a staircase, and Sprint 1's line sits flat along the axis for most of its window because the stories were estimated and moved on the final day. The *shape* of these lines reflects how the simulation was run, not how a team worked.

What the charts do legitimately show:

Scope is visible and correct. Sprint 1 opens at 42 points and Sprint 2 at 49 — the commitments match the plan, and the scope-changes log beneath each chart lists every estimate as it entered the sprint.

Sprint 2's remaining line reaches zero without all the work being done. 13 of its 49 points were not completed; they left the sprint as a scope reduction at close rather than as delivered work. This is the single most important thing to understand when reading a burndown: *a line reaching zero does not by itself mean the sprint succeeded*. The completion figure (36 of 49) and the scope log tell the real story, and they disagree with the shape of the line.

Velocity across the two sprints averages 39 points (42 then 36). That is a more honest planning figure than either sprint alone, and it is only visible because Sprint 2 was allowed to miss. Two data points still do not establish a reliable velocity — a real team needs several sprints — but the direction is already useful: Sprint 2 was over-committed by roughly 10 points.

Large stories hide risk. Story 1.2 at 13 points was a third of Sprint 1 in a single card. Had it slipped, a third of the sprint would have gone with it, and the burndown would have given no warning until the end. The handout names this pattern directly: a sharp drop suggests work that was not estimated accurately or not broken down properly. That argues for splitting the ETA engine — for example into distance-based ETA and traffic-speed modelling.

The gap between guideline and actual is the whole point. In a real project, the size and direction of that gap across several sprints is what tells a team whether it is over-committing, under-committing or estimating well. That feedback loop is the reason the ceremony exists — and it is exactly what a one-session simulation cannot produce.

Key takeaways

1. Story points measure **relative uncertainty**, not hours — which is why the Fibonacci gaps are useful rather than arbitrary.
2. Points do not capture **dependencies**. Sprint 2 missed on sequencing, not on size.
3. Prioritisation should follow **dependency and user impact**, and it worked here: only Medium-priority work slipped.
4. A burndown reaching zero is **not** proof of success — check the completion count and the scope-change log.
5. A 13-point story is a signal to **split**, not a badge of difficulty.